LAGRANGE METHOD OF UNDETERMINED MULTIPLIERS

Till now we have considered the method of optimizing the function of two variables without any conditions. But most of the optimizing problem we come across are of different types, where the function is optimized subject to some conditions.

Let us consider the problem of optimizing a function $Z = f\left(x_1, x_{2,\dots}, x_n\right) \dots \dots (1)$

subject to the conditions

$\Phi_1\left(x_1, x_{2,\dots}, x_n\right) = c_1$

$\Phi_2\left(x_1, x_{2,\dots}, x_n\right) = c_2$

....(2)

$\Phi_r\left(x_1, x_{2,\dots}, x_n\right) = c_r$

Where $c_1, c_2, .. c_r$ are all constants.

One of the ways of solving this problem is to solve, r, n -variables from conditions (2) and substituting in (1), which reduces to a function of $(n - r)$ variables, that can be solved by the direct methods. Now let us consider different method, called the Lagrange's method, which gives stationary points of the function f.

Consider the function $g(x_1, x_2, \dots x_n) = \mathrm{f}(x_1, x_2, \dots x_n) + \sum_{i=1}^{r} \lambda_i \, \Phi_{\mathrm{i}}(x_1, x_2, \dots x_n)$ where λ_i, $i = 1,2, \dots r$ are all constants , is called Lagrange's multipliers.

If the conditions defined by (2) are satisfied, then f($x_1, x_2, \dots x_n$) *and* $g(x_1, x_2, \dots x_n)$ attain optimal value together. When optimal values are attained, then $\frac{\partial g}{\partial x_i} = 0$, $i = 1,2,3 \dots n$ *and* $\frac{\partial g}{\partial \lambda} = 0$

$$i.e, \frac{\partial f}{\partial x_i} + \sum_{i=1}^{r} \lambda_i \frac{\partial \Phi_i}{\partial x_i} = 0 \dots \dots \dots \dots (3)$$

Solving for $x_1, x_2, \dots x_n$ and,, $\lambda_1, \lambda_2, \dots \lambda_r$ from (3) and (2), we get the stationary values of the defined by function $f(x_1, x_2, \dots x_n)$ subject to the conditions (2). The sufficient condition for $f(x_1, x_{2,\dots}, x_n)$ to have maxima or minima is that determinant D has all negative or all positive roots respectively, where

$$D = \begin{vmatrix} g_{11} - z & g_{12} & \dots & g_{1n} & \Phi_{11} & \dots & \Phi_{r1} \\ g_{21} & g_{22} - z & \dots & g_{2n} & \Phi_{12} & \dots & \Phi_{r2} \\ . & . & & . & . & . & . \\ g_{n1} & g_{n2} & \dots & g_{nn} - z & \Phi_{1n} & .. & \Phi_{rn} \\ \Phi_{11} & \Phi_{21} & & .. \; \Phi_{r1} & 0 & \dots & 0 \\ . & . & & . \quad . & . & . & . \\ \Phi_{1n} & \Phi_{2n} & & . \quad \Phi_{rn} & 0 & 0 & 0 \end{vmatrix} = 0.$$

$\Phi_{ij} = \frac{\partial \Phi_i}{\partial x_j}$, $i = 1,2, \dots r, j = 1,2 \dots n$

$g_{ij} = \frac{\partial^2 g}{\partial x_i \partial x_j}$, $i = 1,2, \dots n, j = 1,2 \dots r$

Solved Problems:

1.Find the maximum value of $f = 2x_1 + x_2 + 10$ subject to the condition $x_1 + 2{x_2}^2 = 3$

Solution: *Let* $g(x_1, x_2) = f(x_1, x_2) + \lambda\Phi_1(x_1, x_2) = 2x_1 + x_2 + 10 + \lambda\ (x_1 + 2{x_2}^2 - 3)$

The necessary condition for *f* is, $\frac{\partial g}{\partial x_1} = 0, \frac{\partial g}{\partial x_2} = 0$ and $\frac{\partial g}{\partial \lambda} = 0$

That is, $2 + \lambda = 0, 1 + 4\lambda x_2 = 0,\ x_1 + 2{x_2}^2 - 3 = 0$

On solving we get, stationary values of *f* is

$$(x_1, x_2) = \left(\frac{95}{32}, \frac{1}{8}\right), \lambda = -2$$

$$g_{11} = \frac{\partial^2 g}{\partial {x_1}^2} = 0, g_{12} = \frac{\partial^2 g}{\partial x_1 \partial x_2} = 0 = \frac{\partial^2 g}{\partial x_2 \partial x_1} = g_{21},\ \ g_{22} = \frac{\partial^2 g}{\partial {x_2}^2} = -8,$$

$$\Phi_{11} = \frac{\partial \Phi_1}{\partial x_1} = 1, \Phi_{12} = \frac{\partial \Phi_1}{\partial x_2} = 0.5.$$

Then, the sufficient condition for *f* is

$$D = \begin{vmatrix} g_{11} - z & g_{12} & \Phi_{11} \\ g_{21} & g_{22} - z & \Phi_{12} \\ \Phi_{11} & \Phi_{12} & 0 \end{vmatrix} = \begin{vmatrix} -z & 0 & 1 \\ 0 & -8 - z & 0.5 \\ 1 & 0.5 & 0 \end{vmatrix} = 0$$

$\Rightarrow 1.25z + 8 = 0$

$\Rightarrow z = -6.4$ are the negative roots.

$Hence, f\ has\ maximum\ value$ and is equal to $f_{max} = 6.0625$

2.Find the minimum value of $f = {x_1}^2 + {x_2}^2 + {x_3}^2$ subject to the condition $x_1x_2x_3 = 8$.

Solution: Let $g(x_1, x_2, x_3) = f(x_1, x_2, x_3) + \lambda\Phi_1(x_1, x_2, x_3) = {x_1}^2 + {x_2}^2 + {x_3}^2 + \lambda(x_1x_2x_3 - 8)$.

The necessary condition for *f* is, $\frac{\partial g}{\partial x_1} = 0,\ \frac{\partial g}{\partial x_2} = 0,\ \frac{\partial g}{\partial x_3} = 0$ and $\frac{\partial g}{\partial \lambda} = 0$

That is, $2x_1 + \lambda x_2x_3 = 0, 2x_2 + \lambda x_1x_3 = 0, 2x_3 + \lambda x_1x_2 = 0,\ \ x_1x_2x_3 = 8$

On solving we get, stationary values of *f* is $i)\ (x_1, x_2, x_3) = (2, 2, 2)$ and $\lambda = -1$

$ii)\ (x_1, x_2, x_3) = (-2, -2, 2)$ and $\lambda = -1$

$iii)\ (x_1, x_2, x_3) = (-2,\ 2, -2)$ and $\lambda = -1$

$iv)\ (x_1, x_2, x_3) = (2, -2, -2)$ and $\lambda = -1$

$v)\ (x_1, x_2, x_3) = (-2, -2, -2)$ and $\lambda = 1$

We discard *v)* case. Since $(x_1, x_2, x_3) = (-2, -2, -2)$ not satisfy the condition $x_1x_2x_3 = 8$.

For case i),

$$g_{11} = \frac{\partial^2 g}{\partial {x_1}^2} = 2, g_{12} = \frac{\partial^2 g}{\partial x_1 \partial x_2} = -2 = \frac{\partial^2 g}{\partial x_2 \partial x_1} = g_{21},\ g_{13} = \frac{\partial^2 g}{\partial x_1 \partial x_3} = -2 = \frac{\partial^2 g}{\partial x_3 \partial x_1} = g_{31},$$

$g_{22} = \frac{\partial^2 g}{\partial {x_2}^2} = 2$, $g_{33} = \frac{\partial^2 g}{\partial {x_3}^2} = 2$, $g_{23} = \frac{\partial^2 g}{\partial x_2 \partial x_3} = -2 = \frac{\partial^2 g}{\partial x_3 \partial x_2} = g_{32}$, $\Phi_{11} = \frac{\partial \Phi_1}{\partial x_1} = 4$,

$\Phi_{12} = \frac{\partial \Phi_1}{\partial x_2} = 4, \Phi_{13} = \frac{\partial \Phi_1}{\partial x_3} = 4$.

Then, the sufficient condition for f is

$$D = \begin{vmatrix} g_{11} - z & g_{12} & g_{13} & \Phi_{11} \\ g_{21} & g_{22} - z & g_{23} & \Phi_{12} \\ g_{31} & g_{32} & g_{33} - z & \Phi_{13} \\ \Phi_{11} & \Phi_{12} & \Phi_{13} & 0 \end{vmatrix} = \begin{vmatrix} 2 - z & -2 & -2 & 4 \\ -2 & 2 - z & -2 & 4 \\ -2 & -2 & 2 - z & 4 \\ 4 & 4 & 4 & 0 \end{vmatrix} = 0$$

$\Rightarrow -48(z-4)^2 = 0$

$\Rightarrow z = 4, 4$ are the positive roots.

Similarly, we can find D at other points mentioned in ii) to iv) we get, the same roots.

$Hence, f\ has\ minimum\ value$ and is equal to f_{min} =12.

3. Find the maximum value of $f = x_1 {x_2}^2 {x_3}^3$ subject to the condition $x_1 + x_2 + x_3 = 24$.

Solution: Let $g(x_1, x_2, x_3) = f(x_1, x_2, x_3) + \lambda \Phi_1(x_1, x_2, x_3) = x_1 {x_2}^2 {x_3}^3 + \lambda (x_1 + x_2 + x_3 - 24)$.

The necessary condition for f is, $\frac{\partial g}{\partial x_1} = 0$, $\frac{\partial g}{\partial x_2} = 0$, $\frac{\partial g}{\partial x_3} = 0$ and $\frac{\partial g}{\partial \lambda} = 0$

That is, ${x_2}^2 {x_3}^3 + \lambda = 0$, $2x_1 x_2 {x_3}^3 + \lambda = 0$, $3x_1 {x_2}^2 {x_3}^2 + \lambda = 0$, $x_1 + x_2 + x_3 = 24$

On solving we get, stationary values of f is $(x_1, x_2, x_3) = (4, 8, 12)$ and $\lambda = -110592$

$g_{11} = \frac{\partial^2 g}{\partial {x_1}^2} = 0, g_{12} = \frac{\partial^2 g}{\partial x_1 \partial x_2} = 27648 = \frac{\partial^2 g}{\partial x_2 \partial x_1} = g_{21}$, $g_{31} = \frac{\partial^2 g}{\partial x_1 \partial x_3} = 27648 = \frac{\partial^2 g}{\partial x_3 \partial x_1} = g_{31}$, $g_{22} = \frac{\partial^2 g}{\partial {x_2}^2} = 13824$, $g_{33} = \frac{\partial^2 g}{\partial {x_3}^2} = 18432$, $g_{23} = \frac{\partial^2 g}{\partial x_2 \partial x_3} = 27648 = \frac{\partial^2 g}{\partial x_3 \partial x_2} = g_{32}$, $\Phi_{11} = \frac{\partial \Phi_1}{\partial x_1} = 1, \Phi_{12} = \frac{\partial \Phi_1}{\partial x_2} = 1, \Phi_{13} = \frac{\partial \Phi_1}{\partial x_3} = 1$.

Then, the sufficient condition for f is

$$D = \begin{vmatrix} g_{11} - z & g_{12} & g_{13} & \Phi_{11} \\ g_{21} & g_{22} - z & g_{23} & \Phi_{12} \\ g_{31} & g_{32} & g_{33} - z & \Phi_{13} \\ \Phi_{11} & \Phi_{12} & \Phi_{13} & 0 \end{vmatrix} = \begin{vmatrix} -z & 27648 & 27648 & 1 \\ 27648 & 13824 - z & 27648 & 1 \\ 27648 & 27648 & 18432 - z & 1 \\ 1 & 1 & 1 & 0 \end{vmatrix} = 0$$

$\Rightarrow 3z^2 + 101376z + 764411904 = 0$

$\Rightarrow z = -11357.9, -22434.1$ are the negative roots.

$Hence, f\ has\ maximum\ value$ and is equal to f_{max} =442368.

4. The temperature at any point (x_1, x_2, x_3) in space is $f = 400 x_1 x_2 {x_3}^2$. Find the highest temperature on the surface of the unit sphere ${x_1}^2 + {x_2}^2 + {x_3}^2 = 1$

Solution: Let $g(x_1, x_2, x_3) = f(x_1, x_2, x_3) + \lambda \Phi_1(x_1, x_2, x_3) = 400 x_1 x_2 {x_3}^2 + \lambda ({x_1}^2 + {x_2}^2 + {x_3}^2 - 1)$.

The necessary condition for f is, $\frac{\partial g}{\partial x_1} = 0$, $\frac{\partial g}{\partial x_2} = 0$, $\frac{\partial g}{\partial x_3} = 0$ and $\frac{\partial g}{\partial \lambda} = 0$

That is, $400\, x_2{x_3}^2 + 2\lambda x_1 = 0$, $400x_1{x_3}^2 + 2\lambda x_2 = 0$, $800x_1x_2x_3 + 2\lambda x_3 = 0, {x_1}^2 + {x_2}^2 + {x_3}^2 = 1$

On solving we get, stationary values of f is $(x_1, x_2, x_3) = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}\right)$ and $\lambda = -100$.

$g_{11} = \frac{\partial^2 g}{\partial {x_1}^2} = -200, g_{12} = \frac{\partial^2 g}{\partial x_1 \partial x_2} = 200 = \frac{\partial^2 g}{\partial x_2 \partial x_1} = g_{21},$

$g_{13} = \frac{\partial^2 g}{\partial x_1 \partial x_3} = \frac{400}{\sqrt{2}} = \frac{\partial^2 g}{\partial x_3 \partial x_1} = g_{31},\ g_{22} = \frac{\partial^2 g}{\partial {x_2}^2} = -200\,, g_{33} = \frac{\partial^2 g}{\partial {x_3}^2} = 0,$

$g_{23} = \frac{\partial^2 g}{\partial x_2 \partial x_3} = \frac{400}{\sqrt{2}} = \frac{\partial^2 g}{\partial x_3 \partial x_2} = g_{32}\,,\ \Phi_{11} = \frac{\partial \Phi_1}{\partial x_1} = 1, \Phi_{12} = \frac{\partial \Phi_1}{\partial x_2} = 1, \Phi_{13} = \frac{\partial \Phi_1}{\partial x_3} = 1\,.$

Then, the sufficient condition for f is

$$D = \begin{vmatrix} g_{11} - z & g_{12} & g_{13} & \Phi_{11} \\ g_{21} & g_{22} - z & g_{23} & \Phi_{12} \\ g_{31} & g_{32} & g_{33} - z & \Phi_{13} \\ \Phi_{11} & \Phi_{12} & \Phi_{13} & 0 \end{vmatrix} = \begin{vmatrix} -200 - z & 200 & \frac{400}{\sqrt{2}} & 1 \\ 200 & -200 - z & \frac{400}{\sqrt{2}} & 1 \\ \frac{400}{\sqrt{2}} & \frac{400}{\sqrt{2}} & -z & 1 \\ 1 & 1 & 1 & 0 \end{vmatrix} = 0$$

$\Rightarrow -(z + 400)\left(3z + 800\sqrt{2}\right) = 0$

$\Rightarrow z = -400, -800\frac{\sqrt{2}}{3}$ are the negative roots.

$Hence, f\ has\ maximum\ value$ and is equal to $f_{max} = 50$

5. Find the minimum value of $f = x_1 + x_2$ subject to the condition ${x_1}^2 + {x_2}^2 = 1$

Solution: Let $g(x_1, x_2) = \mathrm{f}(x_1, x_2) + \lambda\Phi_1(x_1, x_2) = x_1 + x_2 + \lambda\,({x_1}^2 + {x_2}^2 - 1)$.

The necessary condition for f is, $\frac{\partial \mathrm{g}}{\partial x_1} = 0, \frac{\partial \mathrm{g}}{\partial x_2} = 0$ and $\frac{\partial \mathrm{g}}{\partial \lambda} = 0$

That is, $1 + 2\lambda x_1 = 0, 1 + 2\lambda x_2 = 0, {x_1}^2 + {x_2}^2 - 1 = 0$

On solving we get, stationary values of f is

$i)(x_1, x_2) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \lambda = -\frac{1}{\sqrt{2}}\ \ and\ ii)(x_1, x_2) = \left(\frac{-1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right), \lambda = \frac{1}{\sqrt{2}}$

$g_{11} = \frac{\partial^2 g}{\partial {x_1}^2} = 2\lambda, g_{12} = \frac{\partial^2 g}{\partial x_1 \partial x_2} = 0 = \frac{\partial^2 g}{\partial x_2 \partial x_1} = g_{21},\ g_{22} = \frac{\partial^2 g}{\partial {x_2}^2} = 2\lambda\,,\ \Phi_{11} = \frac{\partial \Phi_1}{\partial x_1} = 2x_1\,,$

$\Phi_{12} = \frac{\partial \Phi_1}{\partial x_2} = 2x_2\,.$

Then, the sufficient condition for f is

$$i)\ D = \begin{vmatrix} g_{11} - z & g_{12} & \Phi_{11} \\ g_{21} & g_{22} - z & \Phi_{12} \\ \Phi_{11} & \Phi_{12} & 0 \end{vmatrix} = \begin{vmatrix} -\sqrt{2} - z & 0 & \sqrt{2} \\ 0 & -\sqrt{2} - z & \sqrt{2} \\ \sqrt{2} & \sqrt{2} & 0 \end{vmatrix} = 0$$

$\Rightarrow 4z + 2\sqrt{2} = 0$

$\Rightarrow z = -\sqrt{2}$ is the negative root.

$Hence, f\ has\ maximum\ value$ and is equal to $f_{max} = \sqrt{2}$

$$ii)\ D = \begin{vmatrix} g_{11} - z & g_{12} & \Phi_{11} \\ g_{21} & g_{22} - z & \Phi_{12} \\ \Phi_{11} & \Phi_{12} & 0 \end{vmatrix} = \begin{vmatrix} \sqrt{2} - z & 0 & -\sqrt{2} \\ 0 & \sqrt{2} - z & -\sqrt{2} \\ -\sqrt{2} & -\sqrt{2} & 0 \end{vmatrix} = 0$$

$\Rightarrow 4z - 4\sqrt{2} = 0$

$\Rightarrow z = \sqrt{2}$ is the positive root.

$Hence, f\ has\ minimum\ value$ and is equal to $f_{min} = -\sqrt{2}$

Problems:

1. Find the extreme values of the function $f(x, y, z) = \sin x \sin y \sin z$ where x, y, z are the angles of a triangle.
2. Find the largest rectangular parallelepiped inscribed in an ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.
3. Prove that of all rectangular parallelepiped of the same volume, the cube has the least surface.
4. Find the minimum value of $x^2 + y^2 + z^2$ subject to the condition
   i) $yz + xy + xz = 3a^2$
   ii) $x + y + z = 3a$
5. Given $x + y + z = a$, find the maximum value of $x^m y^n z^p$.
6. A rectangular box open at the top have a volume of $32\ m^3$. Find the dimensions of the box requiring least material for construction.

Reference: 1. SINGIRESU S. RAO, Engineering Optimization Theory and Practice, Fourth Edition, 2009, John Wiley & Sons, Inc.